Key Derivations

A worked companion to the Advanced & Quantum Optics course

Part II Physics — University of Cambridge

The derivations that form the spine of the course, worked through in full with every intermediate step shown and the physical reasoning made explicit. They are organised into three tiers by importance: Tier 1 (foundational, examined most reliably), Tier 2 (important and rich in transferable technique), and Tier 3 (worth understanding, less often reproduced in full).

Contents

1 Tier 1 — Must Know Cold

1.1 The paraxial Helmholtz equation and the Gaussian beam

The idea

A monochromatic field obeys the Helmholtz equation. A beam is a wave that travels mostly along one axis with a slowly spreading profile. Factor out the fast plane-wave carrier, keep only the slow envelope, and the Helmholtz equation collapses to a simpler "paraxial" equation whose fundamental solution is the Gaussian beam.

Step 1: Start from the Helmholtz equation. A complex field of single frequency ν satisfies

$$\nabla^2 U + k^2 U = 0, \qquad k = \frac{2\pi}{\lambda}.$$

Step 2: Peel off the carrier wave. A paraxial beam travels mainly along z, so write it as a plane wave modulated by an envelope $A(\mathbf{r})$ that varies slowly compared with the wavelength:

$$U(\mathbf{r}) = A(\mathbf{r})\,\mathrm{e}^{-\mathrm{i}kz}.$$

We now substitute this into the Helmholtz equation. The transverse Laplacian acts only on A; the z-derivatives use the product rule:

$$\frac{\partial^2 U}{\partial z^2} = \left(\frac{\partial^2 A}{\partial z^2} - 2\mathrm{i}k\frac{\partial A}{\partial z} - k^2 A\right)\mathrm{e}^{-\mathrm{i}kz}.$$

Adding $\nabla_T^2 A\,\mathrm{e}^{-\mathrm{i}kz}$ (the transverse part) and $k^2 U$, the $-k^2 A$ and $+k^2 A$ terms cancel, leaving

$$\nabla_T^2 A + \frac{\partial^2 A}{\partial z^2} - 2\mathrm{i}k\frac{\partial A}{\partial z} = 0.$$

Step 3: The slowly varying envelope approximation. "Slowly varying" means A changes little over one wavelength, so its second axial derivative is negligible beside the first:

$$\left|\frac{\partial^2 A}{\partial z^2}\right| \ll \left|k\frac{\partial A}{\partial z}\right|.$$

Dropping $\partial_z^2 A$ gives the

Paraxial Helmholtz equation

$$\nabla_T^2 A - 2\mathrm{i}k\,\frac{\partial A}{\partial z} = 0.$$

This has the same form as the 2D time-dependent Schrödinger equation, with z playing the role of time. The Gaussian beam is its "ground state".

Step 4: The paraboloidal-wave guess and the q-parameter. A spherical wave from a point source, paraxially expanded, has the form $A \propto (1/q)\exp[-\mathrm{i}k\rho^2/2q]$ where $\rho^2 = x^2 + y^2$. The Gaussian beam is obtained by allowing the "distance" q to be *complex*. Substituting the trial solution

$$A(\mathbf{r}) = \frac{A_1}{q(z)}\exp\left[-\mathrm{i}k\frac{\rho^2}{2q(z)}\right]$$

into the paraxial equation and matching terms order by order in ρ shows it is a solution provided

$$\frac{\mathrm{d}q}{\mathrm{d}z} = 1 \qquad \Longrightarrow \qquad q(z) = z + q_0.$$

Choosing the integration constant as $q_0 = \mathrm{i}z_0$ (pure imaginary) is what makes the beam confined rather than diverging.

Step 5: Extract the physical beam parameters. Split the reciprocal of q into real and imaginary parts. With $q = z + \mathrm{i}z_0$,

$$\frac{1}{q} = \frac{1}{z+\mathrm{i}z_0} = \frac{z-\mathrm{i}z_0}{z^2+z_0^2} = \underbrace{\frac{z}{z^2+z_0^2}}_{\equiv 1/R(z)} - \mathrm{i}\underbrace{\frac{z_0}{z^2+z_0^2}}_{\equiv \lambda/\pi W^2(z)} .$$

The convention for splitting $1/q$ is the standard one,

$$\frac{1}{q(z)} = \frac{1}{R(z)} - \mathrm{i}\frac{\lambda}{\pi W^2(z)},$$

because then the exponent $-\mathrm{i}k\rho^2/2q$ contains a real Gaussian factor $\exp[-\rho^2/W^2]$ (the beam profile) and an imaginary factor $\exp[-\mathrm{i}k\rho^2/2R]$ (the spherical wavefront curvature). Reading off the two pieces:

$$\boxed{R(z) = z\left[1+\left(\frac{z_0}{z}\right)^2\right]}, \qquad \boxed{W(z) = W_0\sqrt{1+\left(\frac{z}{z_0}\right)^2}}, \qquad W_0 = \sqrt{\frac{\lambda z_0}{\pi}}.$$

The width is minimal ($W = W_0$, the *waist*) at $z = 0$ and grows hyperbolically; the wavefront is flat at the waist ($R \to \infty$), most strongly curved at $z = z_0$, and flattens again far away ($R \to z$, a diverging spherical wave).

Step 6: The Gouy phase. Writing $q = z + \mathrm{i}z_0$ in amplitude–phase form, the prefactor $1/q$ contributes a slowly varying phase $\zeta(z) = \arctan(z/z_0)$ that runs from $-\pi/2$ to $+\pi/2$ across the focus. Including it, the full field is

$$U(\mathbf{r}) = A_0\frac{W_0}{W(z)}\exp\left[-\frac{\rho^2}{W^2(z)}\right]\exp\left[-\mathrm{i}kz - \mathrm{i}\frac{k\rho^2}{2R(z)} + \mathrm{i}\zeta(z)\right].$$

Why it matters

The single complex number q carries the entire beam geometry — width and curvature together. Because q transforms through any optical system by the same ABCD matrix that propagates rays ($q_2 = (Aq_1 + B)/(Cq_1 + D)$), designing laser-beam optics reduces to matrix multiplication. This derivation is the spine of beam optics, resonator modes, and fibre coupling.

1.2 The interference law and fringe visibility

The idea

When two fields overlap, the detected intensity is not just the sum of the two intensities — there is a cross term. Ensemble-averaging that cross term introduces the degree of coherence, and the contrast of the resulting fringes measures it directly.

Step 1: Superpose and expand. Two fields U_1, U_2 combine to give total field $U = U_1 + U_2$. The intensity is the (ensemble- or time-) average of $|U|^2$:

$$I = \left\langle |U_1+U_2|^2\right\rangle = \left\langle |U_1|^2\right\rangle + \left\langle |U_2|^2\right\rangle + \langle U_1^* U_2\rangle + \langle U_1 U_2^*\rangle .$$

The first two terms are the individual intensities I_1, I_2. The last two are complex conjugates of one another, so their sum is twice the real part:

$$I = I_1 + I_2 + 2\,\mathrm{Re}\,\langle U_1^* U_2\rangle .$$

Step 2: Introduce the degree of coherence. Define the normalised cross-correlation

$$g_{12} = \frac{\langle U_1^* U_2 \rangle}{\sqrt{I_1 I_2}} = |g_{12}|\, \mathrm{e}^{\mathrm{i}\varphi},$$

so that $\langle U_1^* U_2 \rangle = \sqrt{I_1 I_2}\, |g_{12}| \mathrm{e}^{\mathrm{i}\varphi}$. Then

Interference law

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\, |g_{12}| \cos\varphi.$$

Here φ contains both the optical path-difference phase and any intrinsic phase of g_{12}; as it sweeps through multiples of 2π the intensity oscillates — these are the fringes.

Step 3: Visibility. The fringe maxima and minima occur at $\cos\varphi = \pm 1$:

$$I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}\, |g_{12}|, \qquad I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}\, |g_{12}|.$$

The visibility (Michelson contrast) is therefore

$$\mathcal{V} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}\, |g_{12}|.$$

For equal intensities $I_1 = I_2$ this collapses to the clean result

$$\boxed{\mathcal{V} = |g_{12}|}.$$

Why it matters

Measuring fringe contrast measures the modulus of the degree of coherence directly — no absolute calibration needed. Temporal coherence ($g(\tau)$) underlies the Michelson interferogram and OCT; spatial coherence (g_{12} between two points) underlies the stellar interferometer and the van Cittert–Zernike theorem.

1.3 Second-harmonic generation, rectification, and why $\chi^{(2)} = 0$ in centrosymmetric media

The idea

Drive a medium whose polarisation response is slightly nonlinear with a single-frequency field. Squaring a cosine produces a DC term and a doubled-frequency term — optical rectification and second-harmonic generation. A parity argument then shows the whole effect vanishes if the medium has inversion symmetry.

Step 1: The nonlinear polarisation. Expand the polarisation in powers of the field:

$$P = \epsilon_0 \Big(\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \cdots \Big).$$

Keep the second-order term and drive with a monochromatic field $E(t) = E_0 \cos(\omega t)$.

Step 2: Square the cosine.

$$P^{(2)}(t) = \epsilon_0 \chi^{(2)} E_0^2 \cos^2(\omega t) = \epsilon_0 \chi^{(2)} E_0^2 \cdot \tfrac{1}{2}\big[1 + \cos(2\omega t)\big].$$

The bracket splits into two physically distinct pieces:

Two second-order processes

$$P^{(2)}(t) = \underbrace{\tfrac{1}{2}\epsilon_0 \chi^{(2)} E_0^2}_{\text{DC: optical rectification}} + \underbrace{\tfrac{1}{2}\epsilon_0 \chi^{(2)} E_0^2 \cos(2\omega t)}_{\text{oscillates at } 2\omega\text{: SHG}}.$$

A static polarisation appears (optical rectification — the basis of THz generation), and a polarisation oscillating at 2ω appears, which radiates the second harmonic.

Step 3: The parity (symmetry) argument. Suppose the medium is *centrosymmetric* — it looks the same under inversion of all coordinates, $\mathbf{r} \to -\mathbf{r}$. Under this operation both the field and the polarisation, being polar vectors, reverse sign: $E \to -E$ and $P \to -P$. Apply this to the second-order relation $P^{(2)} = \epsilon_0 \chi^{(2)} E^2$:

$$\text{LHS} \to -P^{(2)}, \qquad \text{RHS} \to \epsilon_0 \chi^{(2)} (-E)^2 = \epsilon_0 \chi^{(2)} E^2 = +P^{(2)}.$$

Consistency demands $-P^{(2)} = +P^{(2)}$, i.e. $P^{(2)} = 0$ for all E, which forces

$$\boxed{\chi^{(2)} = 0 \quad \text{(centrosymmetric media)}}.$$

The argument works for every even order; odd orders such as $\chi^{(3)} E^3$ are odd under $E \to -E$ and therefore survive in any medium.

Why it matters

This is the single most important selection rule in nonlinear optics. It explains why second-harmonic crystals must be non-centrosymmetric (quartz, KDP, BBO, $LiNbO_3$) while glasses and liquids — centrosymmetric on average — show only third-order effects such as the Kerr effect. The same logic, expressed through the anharmonic-oscillator potential, says $\chi^{(2)}$ requires an asymmetric (cubic) term in the binding potential.

1.4 Field quantisation: the commutator and the Hamiltonian

The idea

A single mode of the electromagnetic field oscillates exactly like a harmonic oscillator. Promoting the oscillator's position and momentum to operators, and repackaging them into raising and lowering combinations, gives the photon ladder: each quantum of excitation is one photon.

Step 1: The mode is a harmonic oscillator. A single field mode has energy with the same structure as a unit-mass oscillator, $\hat{H} = \frac{1}{2}(\hat{p}^2 + \omega^2\hat{q}^2)$, where the canonical pair $\hat{q}, \hat{p}$ (built from the field quadratures) obeys $[\hat{q}, \hat{p}] = \mathrm{i}\hbar$.

Step 2: Define ladder operators.

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega}}\left(\omega\hat{q} + \mathrm{i}\hat{p}\right), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar\omega}}\left(\omega\hat{q} - \mathrm{i}\hat{p}\right).$$

Step 3: Compute the commutator.

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2\hbar\omega}\left[\omega\hat{q} + \mathrm{i}\hat{p},\ \omega\hat{q} - \mathrm{i}\hat{p}\right].$$

Expand the bracket. The $[\hat{q}, \hat{q}]$ and $[\hat{p}, \hat{p}]$ terms vanish, leaving the cross terms:

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2\hbar\omega}\Big(-\mathrm{i}\omega[\hat{q}, \hat{p}] + \mathrm{i}\omega[\hat{p}, \hat{q}]\Big).$$

Using $[\hat{q}, \hat{p}] = \mathrm{i}\hbar$ and $[\hat{p}, \hat{q}] = -\mathrm{i}\hbar$:

$$[\hat{a}, \hat{a}^\dagger] = \frac{1}{2\hbar\omega}\big(-\mathrm{i}\omega \cdot \mathrm{i}\hbar + \mathrm{i}\omega \cdot (-\mathrm{i}\hbar)\big) = \frac{1}{2\hbar\omega}\big(\hbar\omega + \hbar\omega\big).$$

Bosonic commutation relation

$$[\hat{a}, \hat{a}^\dagger] = 1.$$

Step 4: Rewrite the Hamiltonian. Invert the definitions:

$$\hat{q} = \sqrt{\frac{\hbar}{2\omega}}\,(\hat{a}^\dagger + \hat{a}), \qquad \hat{p} = \mathrm{i}\sqrt{\frac{\hbar\omega}{2}}\,(\hat{a}^\dagger - \hat{a}).$$

Then

$$\hat{p}^2 = -\frac{\hbar\omega}{2}(\hat{a}^\dagger - \hat{a})^2, \qquad \omega^2\hat{q}^2 = \frac{\hbar\omega}{2}(\hat{a}^\dagger + \hat{a})^2.$$

Add them. Expanding the squares, the $\hat{a}^{\dagger 2}$ and $\hat{a}^2$ terms cancel between the two expressions, leaving only the cross terms:

$$\hat{H} = \tfrac{1}{2}(\hat{p}^2 + \omega^2\hat{q}^2) = \frac{\hbar\omega}{2}\big(\hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger\big).$$

Finally use $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ (from Step 3) to normal-order:

Quantised single-mode Hamiltonian

$$\hat{H} = \frac{\hbar\omega}{2}\big(2\hat{a}^\dagger\hat{a} + 1\big) = \hbar\omega\Big(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\Big) = \hbar\omega\big(\hat{n} + \tfrac{1}{2}\big).$$

Step 5: Read off the physics. The number operator $\hat{n} = \hat{a}^\dagger\hat{a}$ has integer eigenvalues $n = 0, 1, 2, \ldots$, so the energies are $E_n = \hbar\omega(n + \frac{1}{2})$: a ladder of equally spaced levels. Each step is one photon of energy $\hbar\omega$; the residual $\frac{1}{2}\hbar\omega$ is the vacuum (zero-point) energy, present even when $n = 0$.

Why it matters

The single relation $[\hat{a}, \hat{a}^\dagger] = 1$ is the entire algebraic engine of quantum optics: it generates the photon-number ladder, fixes the zero-point energy, and underlies every field fluctuation, the coherent state, squeezing, and the Jaynes–Cummings model.

1.5 The coherent state in the Fock basis

The idea

The coherent state is the eigenstate of the annihilation operator — the quantum state that stays unchanged (up to its eigenvalue) when a photon is removed. Solving that eigenvalue equation in the number basis produces a Poissonian superposition of Fock states, the quantum description of laser light.

Step 1: The defining equation.

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle\,, \qquad \alpha \in \mathbb{C}.$$

Expand in the number basis, $|\alpha\rangle = \sum_{n=0}^{\infty} c_n\,|n\rangle$.

Step 2: Apply $\hat{a}$ and match coefficients. Using $\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle$,

$$\hat{a}\,|\alpha\rangle = \sum_{n=1}^{\infty} c_n\sqrt{n}\,|n-1\rangle = \sum_{m=0}^{\infty} c_{m+1}\sqrt{m+1}\,|m\rangle\,,$$

relabelling $m = n-1$. This must equal $\alpha\,|\alpha\rangle = \sum_m \alpha c_m\,|m\rangle$. Matching the coefficient of each $|m\rangle$:

$$c_{m+1}\sqrt{m+1} = \alpha c_m \qquad \Longrightarrow \qquad c_{m+1} = \frac{\alpha}{\sqrt{m+1}}\,c_m.$$

Step 3: Solve the recursion. Iterating from c_0:

$$c_n = \frac{\alpha^n}{\sqrt{n!}}\,c_0.$$

Step 4: Normalise. Demand $\sum_n |c_n|^2 = 1$:

$$|c_0|^2 \sum_{n=0}^{\infty} \frac{|\alpha|^{2n}}{n!} = |c_0|^2\,\mathrm{e}^{|\alpha|^2} = 1 \qquad \Longrightarrow \qquad c_0 = \mathrm{e}^{-|\alpha|^2/2},$$

using the Taylor series $\sum_n x^n/n! = \mathrm{e}^x$. Hence

Coherent state in the Fock basis

$$|\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,.$$

Step 5: Photon statistics. The probability of detecting n photons is

$$P_n = |c_n|^2 = \mathrm{e}^{-|\alpha|^2}\,\frac{|\alpha|^{2n}}{n!} = \mathrm{e}^{-\bar{n}}\,\frac{\bar{n}^{\,n}}{n!},$$

a *Poisson distribution* with mean $\bar{n} = |\alpha|^2$. For the mean photon number directly,

$$\bar{n} = \langle \hat{n} \rangle = \langle\alpha|\,\hat{a}^\dagger\hat{a}\,|\alpha\rangle = (\alpha^*)(\alpha)\,\langle\alpha|\alpha\rangle = |\alpha|^2.$$

For the variance, use $\hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} = \hat{a}^\dagger(\hat{a}^\dagger\hat{a}+1)\hat{a} = \hat{a}^{\dagger 2}\hat{a}^2 + \hat{a}^\dagger\hat{a}$, so

$$\langle \hat{n}^2 \rangle = |\alpha|^4 + |\alpha|^2 = \bar{n}^2 + \bar{n} \quad \Longrightarrow \quad (\Delta n)^2 = \langle \hat{n}^2 \rangle - \bar{n}^2 = \bar{n}.$$

Thus $\Delta n = \sqrt{\bar{n}}$ and the fractional uncertainty is

$$\frac{\Delta n}{\bar{n}} = \frac{1}{\sqrt{\bar{n}}} = \frac{1}{|\alpha|} \longrightarrow 0 \quad \text{as } |\alpha| \to \infty.$$

Why it matters

A coherent state has a non-zero mean field ($\langle\alpha|\,\hat{E}\,|\alpha\rangle \neq 0$, unlike any Fock state), Poissonian photon statistics, and a relative intensity noise that vanishes in the bright limit. This is exactly the behaviour of an ideal laser beam, and it is the bridge between the quantum and classical descriptions of light. The vacuum is the special case $\alpha = 0$.

2 Tier 2 — Important, Technique-Rich

2.1 The Rabi problem and the optical Bloch equations

The idea

A two-level atom driven by a near-resonant field oscillates coherently between its levels. Going to a frame rotating at the drive frequency and discarding fast counter-rotating terms turns the problem into a static 2×2 Hamiltonian, whose solution is the Rabi oscillation. Adding spontaneous decay turns the state vector into a density matrix obeying the optical Bloch equations.

Step 1: Hamiltonian and state. Levels $|g\rangle, |e\rangle$ split by $\hbar\omega_0$; drive frequency ω; detuning $\Delta = \omega - \omega_0$. Write $|\psi\rangle = c_g |g\rangle + c_e |e\rangle$. In the rotating frame, after the rotating-wave approximation (which drops terms oscillating at $\omega + \omega_0$, valid for $|\Delta| \ll \omega_0$), the effective Hamiltonian is

$$H = \frac{\hbar}{2} \begin{pmatrix} -\Delta & \Omega_0 \\ \Omega_0 & \Delta \end{pmatrix},$$

with $\Omega_0 = dE_0/\hbar$ the resonant Rabi frequency (d the dipole matrix element).

Step 2: Solve on resonance ($\Delta = 0$). Then $H = \frac{\hbar\Omega_0}{2}\sigma_x$. The Schrödinger equation $\mathrm{i}\hbar\dot{\mathbf{c}} = H\mathbf{c}$ gives

$$\mathrm{i}\dot{c}_g = \tfrac{\Omega_0}{2} c_e, \qquad \mathrm{i}\dot{c}_e = \tfrac{\Omega_0}{2} c_g.$$

Differentiate the second and substitute the first:

$$\ddot{c}_e = -\tfrac{\Omega_0^2}{4} c_e.$$

With the atom starting in the ground state ($c_e(0) = 0$, $c_g(0) = 1$):

$$c_e(t) = -\mathrm{i}\sin\left(\frac{\Omega_0 t}{2}\right) \quad \Longrightarrow \quad |c_e(t)|^2 = \sin^2\left(\frac{\Omega_0 t}{2}\right).$$

The excited-state population oscillates fully between 0 and 1 at the Rabi frequency.

Step 3: General detuning. Diagonalising the full H gives eigenvalue splitting $\hbar\Omega/2$ with the *generalised Rabi frequency*

$$\Omega = \sqrt{\Omega_0^2 + \Delta^2}.$$

The solution becomes

Rabi oscillation

$$|c_e(t)|^2 = \frac{\Omega_0^2}{\Omega^2} \sin^2\left(\frac{\Omega t}{2}\right).$$

Detuning makes the oscillation faster ($\Omega > \Omega_0$) but caps its amplitude at $\Omega_0^2/\Omega^2 < 1$: a detuned drive can never fully invert the atom.

Step 4: Add decay — the density matrix. A pure state cannot describe spontaneous emission, so switch to the density matrix $\hat{\rho}$ with components $\rho_{ee}, \rho_{gg}, \rho_{eg}, \rho_{ge}$. Coherent evolution follows $\dot{\hat{\rho}} = -\frac{\mathrm{i}}{\hbar}[H, \hat{\rho}]$; spontaneous emission at rate γ adds decay of the excited population and of the coherences. Defining the Bloch components $u = 2\,\mathrm{Re}\,\rho_{eg}$, $v = -2\,\mathrm{Im}\,\rho_{eg}$, $w = \rho_{ee} - \rho_{gg}$, the result is the

Optical Bloch equations

$$\dot{u} = \Delta v - \frac{\gamma}{2}u, \quad \dot{v} = -\Delta u + \Omega_0 w - \frac{\gamma}{2}v, \quad \dot{w} = -\Omega_0 v - \gamma(w+1).$$

Step 5: Steady state. Set all time-derivatives to zero and solve the linear system. The result for the inversion is

$$w_{\rm st} = -\frac{1}{1+S}, \qquad S = \frac{S_0}{1+(2\Delta/\gamma)^2}, \qquad S_0 = \frac{2\Omega_0^2}{\gamma^2},$$

where S is the saturation parameter. At low intensity $w_{\rm st} \to -1$ (atom in ground state); at high intensity $w_{\rm st} \to 0$ (saturation — the population never exceeds $\frac{1}{2}$ because spontaneous emission keeps depleting it). The photon scattering rate is $\Gamma_{\rm sc} = \frac{\gamma}{2}\, S/(1+S)$.

Why it matters

The coherent part (u, v, w rotating) is the Bloch-sphere picture: the state vector rotates about an axis $\propto (\Omega_0, 0, -\Delta)$; the decay terms shrink it inward toward the steady state. This single framework underlies Rabi flopping, π-pulses, Ramsey and spin-echo sequences, laser cooling, and saturation spectroscopy.

2.2 Phase mismatch and the coherence length

The idea

A nonlinear polarisation radiates a new-frequency wave all along the crystal. Unless that wave keeps in step with the polarisation driving it (phase matching), contributions from different depths interfere destructively. The integral over the crystal gives the characteristic sinc^2 conversion curve.

Step 1: Accumulate the generated field. The generated wave amplitude is the sum of contributions $\mathrm{d}E_3$ launched at each depth z, each propagating to the exit with its own phase. The local source is the nonlinear polarisation $\propto E_1 E_2\, \mathrm{e}^{\mathrm{i}(k_1+k_2)z}$, while a free wave at ω_3 propagates as $\mathrm{e}^{\mathrm{i}k_3 z}$. The mismatch between them is

$$\Delta k = k_3 - k_1 - k_2.$$

Integrating the source over a crystal of length L,

$$E_3 \propto \int_0^L \mathrm{e}^{\mathrm{i}\Delta k\, z}\, \mathrm{d}z = \frac{\mathrm{e}^{\mathrm{i}\Delta k L} - 1}{\mathrm{i}\Delta k}.$$

Step 2: Take the intensity.

$$I_3 \propto |E_3|^2 = \left| \frac{\mathrm{e}^{\mathrm{i}\Delta k L} - 1}{\mathrm{i}\Delta k} \right|^2.$$

Write $\mathrm{e}^{\mathrm{i}\Delta kL} - 1 = \mathrm{e}^{\mathrm{i}\Delta kL/2}(\mathrm{e}^{\mathrm{i}\Delta kL/2} - \mathrm{e}^{-\mathrm{i}\Delta kL/2}) = \mathrm{e}^{\mathrm{i}\Delta kL/2} \cdot 2\mathrm{i}\sin(\Delta kL/2)$. Hence

$$|E_3|^2 = \frac{4\sin^2(\Delta kL/2)}{\Delta k^2} = L^2\, \frac{\sin^2(\Delta kL/2)}{(\Delta kL/2)^2} = L^2\, \mathrm{sinc}^2\!\left(\frac{\Delta kL}{2}\right).$$

Conversion efficiency

$$I_3 \propto L^2\, \mathrm{sinc}^2\!\left(\frac{\Delta k\, L}{2}\right).$$

Step 3: The coherence length. The output is maximal at $\Delta k = 0$, where $I_3 \propto L^2$ (coherent growth). For $\Delta k \neq 0$ it first vanishes when $\Delta kL/2 = \pi$, defining the coherence length

$$L_c = \frac{2\pi}{|\Delta k|}.$$

Beyond L_c the generated wave is in antiphase with its source and energy flows back into the fundamental. Quasi-phase matching (periodic poling with period $\Lambda = L_c$) flips the sign of $\chi^{(2)}$ every coherence length, resetting the phase error before it can reverse the power flow.

Why it matters

This is why "more crystal" only helps when phase matched, and why birefringent or quasi-phase matching is essential. The sinc^2 curve — a sharp central peak with rapidly decaying side-lobes — is the quantitative statement of momentum conservation for waves.

2.3 Resonator stability and the free spectral range

The idea

Treat one round trip of a ray as a single ABCD matrix. A ray stays confined only if repeated application of that matrix keeps it bounded — an eigenvalue condition that reduces to a simple inequality in the mirror g-parameters. The wave picture then fixes the resonance frequencies.

Step 1: Round-trip matrix and the stability condition. For a two-mirror resonator (mirrors of radii R_1, R_2, spacing d), one round trip is a product of propagation and mirror matrices with overall form $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$. After N round trips the ray stays bounded only if the eigenvalues of M are complex (pure rotation, $|\lambda| = 1$) rather than real (one > 1, exponential walk-off). The eigenvalues satisfy $\lambda^2 - (A+D)\lambda + 1 = 0$ (since $\det M = 1$), giving

$$\lambda = \frac{A+D}{2} \pm \sqrt{\left(\frac{A+D}{2}\right)^2 - 1}.$$

They are complex iff $|A+D| \le 2$, i.e. $-1 \le \frac{1}{2}(A+D) \le 1$. Evaluating $\frac{1}{2}(A+D)$ in terms of the geometry and defining the g-parameters $g_i = 1 + d/R_i$, this becomes the

Resonator stability condition

$$0 \le g_1 g_2 \le 1, \qquad g_i = 1 + \frac{d}{R_i}.$$

For a symmetric resonator ($R_1 = R_2 = R$) this gives $0 \le d/|R| \le 2$, with planar ($d/|R| = 0$), confocal ($= 1$) and concentric ($= 2$) as the special cases; confocal sits safely in the middle and is most robust to misalignment.

Step 2: Resonance frequencies (wave picture). A standing wave requires the round-trip phase to be a multiple of 2π: $2kd = 2\pi q$ for integer q. With $k = 2\pi\nu/c$,

$$\nu_q = q\,\frac{c}{2d}.$$

Adjacent longitudinal modes are therefore separated by the

Free spectral range

$$\nu_F = \frac{c}{2d}.$$

Step 3: Linewidth and finesse. Including mirror losses, the intracavity intensity versus frequency is an Airy function with peaks of width $\delta\nu$. The ratio of spacing to width is the finesse

$$\mathcal{F} = \frac{\nu_F}{\delta\nu} = \frac{\pi\sqrt{|r|}}{1-|r|},$$

with r the round-trip amplitude reflectivity, and the photon lifetime is $\tau_p = 1/(c\alpha_r)$, related to the linewidth by $\delta\nu = 1/(2\pi\tau_p)$.

Why it matters

The ray result ($0 \le g_1 g_2 \le 1$) and the wave result (a confined Gaussian mode exists, $z_0^2 > 0$) agree exactly — a key consistency check. A resonator is characterised by just two numbers: ν_F from the geometry and $\mathcal{F}$ from the losses.

2.4 The Hong–Ou–Mandel effect

The idea

Send one photon into each input of a 50:50 beam splitter. The amplitude for both to be transmitted and the amplitude for both to be reflected interfere destructively, so the two photons always leave together. There are never coincidences — a purely quantum effect with no classical analogue.

Step 1: Beam-splitter relations. A lossless 50:50 beam splitter relates the input creation operators (1, vacuum port 0) to the outputs $(2, 3)$ by

$$\hat{a}_1^\dagger = \frac{1}{\sqrt{2}}\big(\mathrm{i}\,\hat{a}_2^\dagger + \hat{a}_3^\dagger\big), \qquad \hat{a}_0^\dagger = \frac{1}{\sqrt{2}}\big(\hat{a}_2^\dagger + \mathrm{i}\,\hat{a}_3^\dagger\big),$$

the factors of i enforcing unitarity (energy conservation).

Step 2: Build the two-photon input. One photon in each input port is

$$|1\rangle_0 \, |1\rangle_1 = \hat{a}_0^\dagger \hat{a}_1^\dagger \, |0\rangle \,.$$

Substitute the relations:

$$\hat{a}_0^\dagger \hat{a}_1^\dagger = \frac{1}{2}\big(\hat{a}_2^\dagger + \mathrm{i}\hat{a}_3^\dagger\big)\big(\mathrm{i}\hat{a}_2^\dagger + \hat{a}_3^\dagger\big).$$

Step 3: Expand and simplify.

$$\hat{a}_0^\dagger \hat{a}_1^\dagger = \frac{1}{2}\Big(\mathrm{i}\,\hat{a}_2^{\dagger 2} + \hat{a}_2^\dagger \hat{a}_3^\dagger - \hat{a}_3^\dagger \hat{a}_2^\dagger + \mathrm{i}\,\hat{a}_3^{\dagger 2}\Big).$$

Since creation operators for different modes commute, $\hat{a}_2^\dagger \hat{a}_3^\dagger = \hat{a}_3^\dagger \hat{a}_2^\dagger$, the two cross terms cancel:

$$\hat{a}_0^\dagger \hat{a}_1^\dagger = \frac{\mathrm{i}}{2}\big(\hat{a}_2^{\dagger 2} + \hat{a}_3^{\dagger 2}\big).$$

Step 4: Read off the output state. Using $\hat{a}^{\dagger 2}\,|0\rangle = \sqrt{2}\;|2\rangle$,

HOM output

$$|1\rangle_0 \, |1\rangle_1 \xrightarrow{\text{BS}} \frac{\mathrm{i}}{\sqrt{2}}\big(|2\rangle_2 \, |0\rangle_3 + |0\rangle_2 \, |2\rangle_3\big).$$

Both photons emerge in the *same* output port (either both in 2 or both in 3); the state with one photon in each output has vanished. A coincidence detector therefore registers *nothing* — the "coincidence dip".

Step 5: The physical interference. The amplitude for both photons transmitted and the amplitude for both reflected are equal in magnitude and opposite in sign (because two reflections contribute $\mathrm{i} \times \mathrm{i} = -1$); they cancel. Introducing a relative delay τ between the photons makes them distinguishable, restoring coincidences and tracing out the HOM dip whose depth and width measure their indistinguishability.

Why it matters

HOM is the cleanest demonstration that photon statistics are genuinely non-classical: the effect depends on two-photon amplitude interference, not on field intensities, and the vacuum entering the unused port is essential to the bookkeeping. It is the workhorse test of single-photon indistinguishability in photonic quantum technology.

3 Tier 3 — Worth Understanding

3.1 Waveguide self-consistency condition and mode count

The idea

A guided mode is a wave that reproduces itself after one round trip across the guide and back (two reflections). Demanding that the accumulated transverse phase be a multiple of 2π quantises the allowed bounce angles — these discrete angles are the modes.

Step 1: Self-consistency. In a slab of width d and index n_1, a ray bounces at angle θ to the normal. After one round trip across and back, the transverse phase accumulated is $2k_y d$ with $k_y = n_1 k_0 \cos\theta$, plus any phase shifts φ_r on the two reflections. Self-consistency (the wave reproducing itself) requires

$$2n_1 k_0 d \cos\theta_m - 2\varphi_r = 2\pi m, \qquad m = 0, 1, 2, \ldots$$

Step 2: Mirror waveguide (simplest case). For a planar mirror waveguide the reflection phase is fixed ($\varphi_r = \pi$ for an ideal mirror), and the condition reduces to

$$\sin\theta_m = \frac{m\lambda}{2d},$$

giving a discrete set of bounce angles. The number of guided modes is the number of integers m for which $\sin\theta_m \le 1$:

$$M = \left\lceil \frac{2d}{\lambda} \right\rceil.$$

Step 3: Dielectric waveguide. Here total internal reflection gives an angle-dependent phase shift $\varphi_r(\theta)$ (from the Fresnel equations), so the modes are no longer evenly spaced and there is no overall cutoff — the fundamental $m = 0$ mode always propagates. The number of modes is approximately

$$M \approx \frac{2d}{\lambda_0}\,\mathrm{NA}, \qquad \mathrm{NA} = \sqrt{n_1^2 - n_2^2},$$

with n_2 the cladding index. Single-mode operation requires $M \lesssim 1$.

Why it matters

The single idea — a wave must reproduce itself after a round trip — generates the discrete mode spectrum, exactly as it does for resonators and laser cavities. The only change from the mirror to the dielectric guide is the TIR phase φ_r, which removes the cutoff and uneven spacing.

3.2 Free-space propagation and Fraunhofer as a Fourier transform

The idea

Decompose a field into plane waves. Each travels a distance d by acquiring a phase set by its transverse spatial frequency. In the far field this phase makes the pattern the Fourier transform of the input — which is also why sub-wavelength detail (evanescent waves) never reaches a distant screen.

Step 1: Angular spectrum. Write the input field as a superposition of plane waves with transverse spatial frequencies (ν_x, ν_y):

$$f(x, y) = \iint F(\nu_x, \nu_y)\, \mathrm{e}^{-\mathrm{i}2\pi(\nu_x x + \nu_y y)}\, \mathrm{d}\nu_x\, \mathrm{d}\nu_y.$$

Step 2: Propagate each component. A plane wave with transverse frequencies (ν_x, ν_y) has axial wavevector $k_z = 2\pi\sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2}$. Propagating a distance d multiplies it by $\mathrm{e}^{-\mathrm{i}k_z d}$, so the *transfer function of free space* is

$$H(\nu_x, \nu_y) = \exp\left[-\mathrm{i}2\pi d\sqrt{\lambda^{-2} - \nu_x^2 - \nu_y^2}\right].$$

When $\nu_x^2 + \nu_y^2 > \lambda^{-2}$ the root is imaginary and H becomes a real *decaying* exponential: these evanescent waves carry the sub-wavelength detail but die away within a few wavelengths — the origin of the diffraction limit.

Step 3: Paraxial (Fresnel) limit. For small angles expand the root, $\sqrt{\lambda^{-2} - \nu^2} \approx \lambda^{-1} - \frac{\lambda}{2}\nu^2$, giving a parabolic phase $H \approx H_0 \exp[\mathrm{i}\pi\lambda d(\nu_x^2 + \nu_y^2)]$.

Step 4: Far-field (Fraunhofer) limit. When the Fresnel number $N_F = a^2/\lambda d \ll 1$ (aperture size a), the quadratic phase across the aperture is negligible and the field reduces to

Fraunhofer diffraction

$$g(x, y) \propto F\left(\frac{x}{\lambda d}, \frac{y}{\lambda d}\right).$$

The far-field pattern is the Fourier transform of the input field, evaluated at spatial frequencies $\nu_x = x/\lambda d$. A lens performs the same transform in its focal plane, without needing the large distance d.

Why it matters

Propagation, diffraction and imaging are the same linear operation viewed in the spatial-frequency domain. The evanescent cutoff is the physical reason a classical lens cannot resolve below $\sim \lambda/2$.

3.3 The electro-optic (Pockels) and Kerr index changes

The idea

A nonlinear polarisation involving one optical field and one or two extra fields renders the refractive index controllable — by a DC voltage (Pockels, $\chi^{(2)}$) or by optical intensity (Kerr, $\chi^{(3)}$). Both follow from expanding the effective permittivity to first order in the small non-linear term.

Pockels effect. With one optical field at ω and a static field E_0, the second-order polarisation contributes an extra term to the displacement field,

$$D = \epsilon_0\left[1 + \chi^{(1)} + \chi^{(2)}E_0\right]E.$$

The medium stays linear in E but with an E_0-dependent index:

$$n(E_0) = \sqrt{1 + \chi^{(1)} + \chi^{(2)}E_0} \approx n_0 + \frac{\chi^{(2)}}{2n_0}E_0,$$

expanding the square root because the nonlinear term is small. The index shift is *linear* in the applied voltage — the basis of the Pockels cell modulator. (Requires $\chi^{(2)} \neq 0$, hence a non-centrosymmetric crystal.)

Kerr effect. With a single optical field, the third-order process $\chi^{(3)}(\omega; \omega, \omega, -\omega)$ gives a polarisation $\propto |E|^2E$. Since the intensity is $I = n_0\epsilon_0 c|E|^2$,

$$n(I) = \sqrt{1 + \chi^{(1)} + \frac{\chi^{(3)}}{n_0\epsilon_0 c}I} \approx n_0 + \underbrace{\frac{\chi^{(3)}}{2n_0^2\epsilon_0 c}}_{\equiv\, n_2} I.$$

Index changes

$$n(E_0) = n_0 + \frac{\chi^{(2)}}{2n_0}E_0 \quad \text{(Pockels)}, \qquad n(I) = n_0 + n_2 I \quad \text{(Kerr)}.$$

Consequences of the Kerr effect. A pulse accumulates phase $\varphi = \omega_0 t - k_0 z[n_0 + n_2 I(t)]$, so its instantaneous frequency $\omega(t) = \dot{\varphi} = \omega_0 - k_0 z\, n_2 \dot{I}$ is swept (chirped): red on the rising edge, blue on the falling edge — self-phase modulation. Spatially, the higher index at the intense beam centre acts as a lens — self-focusing. Balanced against dispersion or diffraction, the Kerr effect gives solitons.

Why it matters

Pockels is linear in a DC field (and needs broken inversion symmetry); the optical Kerr effect is linear in intensity (and works in any medium). Distinguishing which field the index depends on is the key to telling them apart. For two distinct beams the cross-phase modulation coefficient is twice the self-phase value.

3.4 Ramsey fringes and the $1/(T\sqrt{N})$ sensitivity

The idea

Two short $\pi/2$ pulses separated by a long free-precession time T let the atomic phase wind at the detuning Δ. Converting that phase back into a population gives fringes in $\cos(\Delta T)$. The narrowness of the fringe ($1/T$) and the shot noise of counting N atoms ($1/\sqrt{N}$) combine to give the frequency sensitivity.

Step 1: First $\pi/2$ pulse. Start at the south pole $(0, 0, -1)$. A resonant $\pi/2$ pulse (rotation about the u-axis) takes the Bloch vector to the equator, $(0, 1, 0)$.

Step 2: Free precession. With the drive off, the vector precesses about the w-axis at the detuning Δ for time T:

$$(u, v, w) = (\sin \Delta T,\ \cos \Delta T,\ 0).$$

Step 3: Second $\pi/2$ pulse. The same rotation about u maps $v \to w$, giving the final inversion

$$w = \cos(\Delta T) \quad \Longrightarrow \quad \rho_{ee} = \frac{1 + \cos \Delta T}{2} = \cos^2\left(\frac{\Delta T}{2}\right).$$

These are the Ramsey fringes: a cosine in ΔT with period $2\pi/T$, so the central fringe has width $\delta\Delta_{\text{fringe}} \sim 1/T$.

Step 4: Shot noise. Each atom yields a binary outcome with excited probability $p = \cos^2(\Delta T/2)$. Averaging N independent atoms, the uncertainty on the estimated probability is the binomial shot noise $\delta p \sim 1/\sqrt{N}$.

Step 5: Propagate to frequency. Convert δp into $\delta\Delta$ via the fringe slope, $\delta\Delta = \delta p/|\mathrm{d}p/\mathrm{d}\Delta|$. The slope is steepest on the side of a fringe, where $|\mathrm{d}p/\mathrm{d}\Delta| \sim T$. Hence

Ramsey frequency sensitivity

$$\delta\Delta \sim \frac{\delta p}{T} \sim \frac{1}{T}\frac{1}{\sqrt{N}}.$$

Why it matters

The two factors are independent: $1/T$ is coherent (longer interrogation, narrower fringe, limited by the coherence time $1/\gamma$ or transit time), while $1/\sqrt{N}$ is statistical (more atoms, better counting). With decay the fringe envelope decays as $\mathrm{e}^{-\gamma T/2}$ — coherences live twice as long as populations — so it washes out for $T \gtrsim 2/\gamma$. This scaling is exactly why atomic clocks use long Ramsey interrogation times.

End of derivations companion. Each derivation is self-contained; read it alongside the corresponding section of the main study handout for the surrounding context and the diagram analysis.